IDEAPAD SEED FUND APPLICATION

ITM University Gwalior — Innovation & Incubation Centre

App ID: app-sample-2026-001 · Version: v1 · Status: SUBMITTED

Target Grant Category: Validation Grant (¹5,000– ¹20,000)

Submitted On (IST): 04 Nov 2026, 04:00 pm IST

Requested Total: ¹1,75,000.00

Canonical Hash: f177f727a1c0...

1. APPLICANT & STARTUP DETAILS

Team / Startup Name: AeroDynamics AI (एयरोडायनामिक्स एआई)
Applicant Full Name: Aarav Sharma (आरव शर्मा)
Enrollment Number: ENR-2026-MECH-042

Program & Department: B.Tech Mechanical Engineering · School of Engineering & Technology
Contact Phone: +91 9876543210
Registered Email: aarav.sharma@ideapad.itm.ac.in

2. GRANT POLICY AGREEMENT ACCEPTANCE

' **Official Terms & Conditions Accepted (Version 1.0)**
Accepted by Aarav Sharma on 04 Nov 2026, 04:00 pm IST
Agreement SHA-256 Text Hash: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855

3. BASIC INFORMATION & CONCEPT

Problem Statement (Problem you are solving):
Smallholder farmers in central India suffer 20–35% yield loss annually from early fungal blight and pest infestation due to late diagnosis. Existing satellite imagery lacks the spatial resolution (10m/pixel) needed for single-plant diagnosis, while commercial multispectral drones remain cost-prohibitive (> ¹8,00,000) for local agricultural co-operatives.

Proposed Solution:
A localized autonomous multispectral aerial mapping platform leveraging edge-AI computer vision. The system captures NDVI and RedEdge reflectance at 2cm/pixel ground resolution to identify crop stress 14 days before visible leaf discoloration, delivering SMS-based actionable remediation maps directly to farmer co-ops.

Effort and Traction So Far:
Fabricated initial proof-of-concept frame; conducted 12 calibration flights across 60 acres in Morena district in partnership with 8 cooperative farmers. Benchmarked NDVI accuracy at 92.4% correlation against ground handheld spectrometers.

4. EXPERIMENT DEFINITION (10 Headings)

1. Objective (Hypothesis being tested):
Validate edge-computed NDVI reflectance stress detection thresholds under varying sunlight angles across 10 distinct soil plots.

2. Problem Statement (Specific technical/market gap):
Atmospheric light variability causes false positives in automated NDVI vegetation indices during midday solar glare.

3. Expected Outcome (TRL-3 proof/validation milestone):
TRL-3 mathematical validation demonstrating <5% variance in edge-calculated vegetative stress indices under fluctuating solar conditions.

4. Methodology (Experiment execution steps):
Daily automated flight grids at 40m AGL across 5 pilot farms during morning, midday, and late afternoon over 8 consecutive weeks.

5. Experimental Setup (Tools, hardware, equipment):
Hexacopter test rig with integrated downwelling light sensor (DLS), calibrated multispectral 4-band camera, and Jetson Nano companion computer.

6. Data to be Collected (Metrics, measurements, observations):
Raw 16-bit GeoTIFF radiance matrices, calibrated reflectance values, ground ambient lux, and ground truth soil moisture readings.

7. Validation Criteria (Pass/fail benchmarks):
Achieve >90% precision in distinguishing nitrogen deficiency from water stress within 48 hours of onset.

8. Risks and Assumptions (Key dependencies & risks):
Inclement monsoon weather mitigation via geo-fenced fail-safe return-to-launch (RTL) protocol and IP54 sensor pod.

9. Timeline (Key stages, schedule & completion targets):
Month 1–2: Optical bench calibration; Month 3–4: 25 farm pilot scans; Month 5–6: Final cooperative report & open validation workshop.

10. Deliverables (Working prototype, validation report, demo):
Bench-tested working multispectral sensor pod, automated flight telemetry logs, 25 verified farm health reports, and TRL-3 demonstration video.

5. ITEMIZED BUDGET BREAKDOWN (Annexure 1)

#	Category	Item Description	Vendor / Source	Unit Cost	Qty	Total (¹)
1	EQUIPMENT	Multispectral 4-Band Optical Sensor	Sentera Technology Inc.	¹110000.00	1	**¹110000.00**
2	CONSUMABLES	Carbon Fiber Lightweight Airframe Propellers (Set of 6)	RoboWings India	¹4000.00	4	**¹16000.00**
3	CONSUMABLES	6S 16000mAh High-Density LiPo Flight Batteries	Tattu Energy	¹9500.00	2	**¹19000.00**
4	TRAVEL	Inter-district Field Transit for Agronomic Trial Sites (Gwalior-Chambal)	State Transit & Fuel	¹15000.00	1	**¹15000.00**
5	CONTINGENCY	Unforeseen Field Spares & Emergency Sensor Enclosure Replacements	Various Approved Vendors	¹15000.00	1	**¹15000.00**

Direct Items Subtotal: ¹175000.00
Contingency (0.0%): ¹0.00
Grand Total: ¹175000.00

6. BUDGET JUSTIFICATIONS (By Item)

Item #1: EQUIPMENT (¹110000.00) — Multispectral 4-Band Optical Sensor
Justification: Standard incubator baseline.
Item #2: CONSUMABLES (¹16000.00) — Carbon Fiber Lightweight Airframe Propellers (Set of 6)
Justification: Standard incubator baseline.
Item #3: CONSUMABLES (¹19000.00) — 6S 16000mAh High-Density LiPo Flight Batteries
Justification: Standard incubator baseline.
Item #4: TRAVEL (¹15000.00) — Inter-district Field Transit for Agronomic Trial Sites (Gwalior-Chambal)
Justification: Standard incubator baseline.
Item #5: CONTINGENCY (¹15000.00) — Unforeseen Field Spares & Emergency Sensor Enclosure Replacements
Justification: Standard incubator baseline.

7. TECHNOLOGY & READINESS LEVELS (Self-Assessment)

Technology Readiness Level (TRL): Level 3: Experimental proof
Manufacturing Readiness Level (MRL): Level 2: Manufacturing concept defined
Integration Readiness Level (IRL): Level 3: Interface proof of concept
Milestone Track Selected: Idea Prototyping and proof of concept (PROTOTYPING)

Track Milestones:

1. [' Done] Optomechanical Sensor Assembly & Laboratory Radiometric Calibration — Bench calibration completed at University Physics Lab
2. [%Ë Pending] Field Testing & Multi-Farm Pilot Data Validation (25 Farms) — Awaiting harvest season flight permits
3. [%Ë Pending] Integrated Prototype Demonstration & Technical Validation Dossier — Scheduled for SEC final review panel

8. TEAM & UPLOADED DOCUMENTS

Founding Team Members:
Aarav Sharma, Neha Gupta

Uploaded Document Registry (Names & Categories Only):
(Note: Identity documents and binary file contents are omitted for privacy and security)

- PAN Card: "founder_pan.pdf" [Member: Aarav Sharma]
- Aadhaar Card: "founder_aadhaar.pdf" [Member: Aarav Sharma]
- PITCH_DECK: "aerodynamics_pitch_deck.pdf"
- Vendor Quotation: "sentera_sensor_quotation.pdf"

9. FINAL DECLARATION & ELECTRONIC SIGNATURE

I hereby declare that all information provided in this application is true, accurate, and complete to the best of my knowledge. The requested seed grant funds will be utilized strictly for the objectives and milestones specified herein in compliance with ITM University IDEAPAD seed fund regulations.

' Electronically Signed by: Aarav Sharma
Signature Timestamp: 04 Nov 2026, 04:00 pm IST